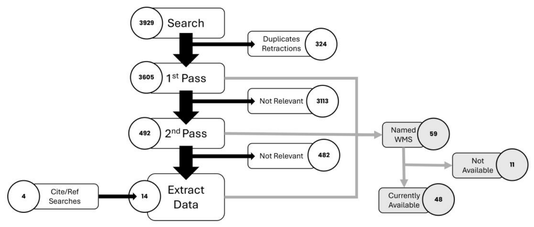
# What Do Biologists Want?

# Finding the Best Workflow Management System for Users

## Scoping Review

Workflow Management Systems (WMS) make it easier for anyone to run a bioinformatics analysis by combining tools into reusable pipelines, but selecting the right WMS can be challenging.

**Flow Chart of Included Papers & WMS**




A scoping review was performed to identify the characteristics of bioinformatics WMS that are most relevant to biologists and to explore how they have previously been evaluated. The aim was to better understand what users should be looking for when choosing a WMS and to create a reusable framework for evaluating these platforms from the user perspective. Papers published since 2018 were included if they evaluated bioinformatics WMS based on criteria relating to the user experience. The most recent platform papers and websites of WMS identified during the search and extraction phases were also included to identify characteristics highlighted by their developers. The review included 14 evaluation papers and identified 48 currently available bioinformatics WMS that were referenced in the literature.

**Three Ways to Evaluate 'Ease of Use'**

| Paper | Definition | Nextflow | Galaxy |
|---|---|---|---|
| Karim et al., 2018 | enables use with little or no training overheads | NA | 1 |
| Marinova & Lazarov, 2019 | effort required to use or share workflows | ★★★★★ | ★★★★★ |
| Wratten et al., 2021 | type of interface (GUI scores 3) | ●●○ | ●●● |

## How Have WMS Been Evaluated?

The scoping review found that the criteria and evaluation processes used by the included papers varied greatly, even for seemingly similar criteria, such as 'Ease of Use'.

Since criteria were not always defined and some papers did not fully explain how they scored the WMS, the evaluations were not usually comparable between papers - and sometimes conflicted.

Evaluations focusing on the needs of users without programming experience favoured Graphical User Interfaces (GUIs) while those aimed at developers preferred text-based interfaces. The limited information on methods meant it would not generally be possible to apply the criteria to other WMS, even though the results quickly became outdated due to changes in availability and updates to the WMS. Clearly defined and easy to reuse criteria are needed to help users choose a WMS.

12 recurring themes relating to user needs emerged from the evaluation criteria and the characteristics highlighted by WMS developers. A **set of reusable criteria** focusing on users from **non-computational backgrounds** was designed and applied to a selection of platforms. The criteria can also help other types of users select the best WMS for their needs.

I'm a developer - I need an open-source WMS that I can test new tools on

I'm a student - I need a free, online WMS that's easy to use without programming

I'm a biologist - I need a WMS that helps me produce FAIR data & run reproducible analyses

I work in industry - we need a secure, scalable WMS that we can customise

I'm an educator - I need an easy to learn WMS that is accessible for all my students

## Shortlisting

Five themes related to user/project-specific requirements and were suggested for use as shortlisting criteria in the evaluation framework: Computing Requirements, Available Functions, Security, Scalability & Cost. Other requirements such as Accessibility will also be important for certain users.

## Evaluation Criteria

Seven themes represented broader user needs and were used to define the evaluation criteria. Most of these themes are scored from 1-5. The first point is assigned for meeting the basic requirements for a WMS. If the platform achieves this, up to four more points can be added for additional benefits. If the platform does not meet the basic requirement, it is unlikely to be worth further consideration.

**SUSTAINABILITY:** How likely is the WMS to continue being available?

**USABILITY:** How easy is it to use for biologists?

**LEARNABILITY:** Are documentation & training materials available?

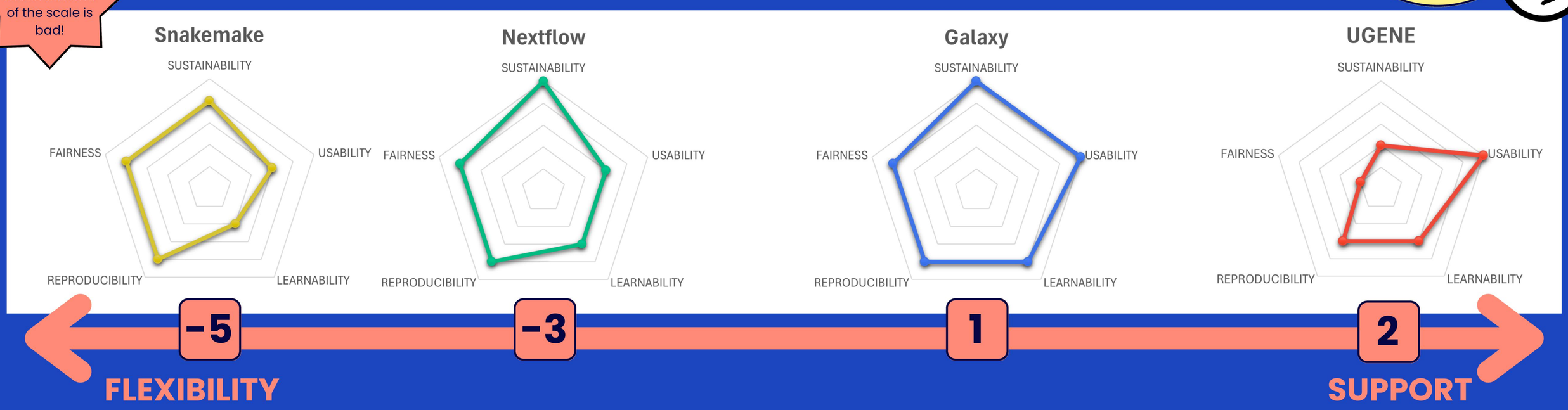
**REPRODUCIBILITY:** Does the WMS enable reproducible research?

**FAIRness:** Are data & workflows FAIR?

The two remaining themes were often scored in opposing directions in the evaluation papers as they reflected the needs of users with different computational skills. These criteria are scored by selecting between five pairs of contrasting characteristics. A score of -1 is assigned if the WMS is more flexible, +1 if it is more supportive, or 0 if it is somewhere in between.

| FLEXIBILITY -1 | SUPPORT +1 |
|---|---|
| Open Access | Managed |
| Additional software required | Ready-to-use |
| Import any tool | Curated selection of tools |
| All parameters open | Guided choice of parameters |
| You're on your own! | Support service on hand! |

Negative scores don't mean this end of the scale is bad!





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**References:**

1. Karim MR, Michel A, Zappa A, Baranov P, Sahay R, Rebholz-Schuhmann D. Improving data workflow systems with cloud services and use of open data for bioinformatics research. Brief Bioinform. 2018 Sep 28;19(5):1035–50.
2. Marinova M, Lazarov V. Comparative analysis of workflow platform in support of in silico oncology. In: AIP Conference Proceedings [Internet]. 2019.
3. Wratten L, Wilm A, Göke J. Reproducible, scalable, and shareable analysis pipelines with bioinformatics workflow managers. Nat Methods. 2021 Oct;18(10):1161–8.